PowerPoint Gesture Assistant

Multimodal Interaction - 3rd Assignment

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Introduction and Project Objective

For this project, we were tasked with developing an assistant to control an app, platform, game, or similar, using an innovative input modality. We implemented a gesture-controlled system for PowerPoint presentations, using a Kinect Sensor to detect user gestures. After recording the gestures in Kinect Studio, we use Kinect Gesture Builder to add the clips to the project correspondent to the gesture in the video and we mark the positive takes. We recorded a clip for each gesture and in each one of those we had, in average, 10 marks. Then we use the resultant built solution in the GesturesModality and through the FusionEngine it communicates with the MMI framework, in order for this one to send the message with the gesture to our C# backend that will finally use the PowerPoint Interop API to interact with the presentation. We also have speech feedback for some of the functionalities and for that we used the WebAssistant.

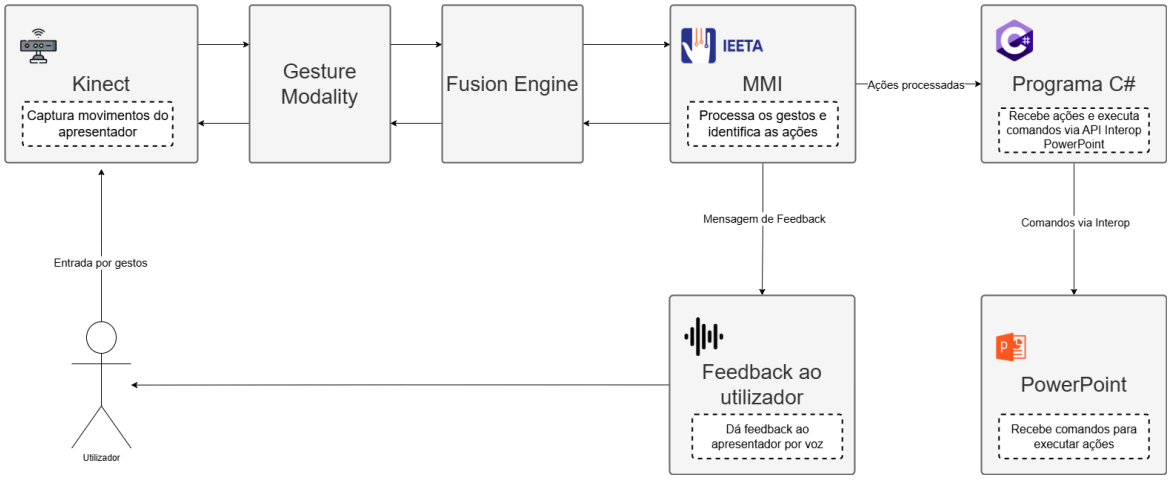


Figure 1: System Architecture

This assistant enhances the flow of a presentation by allowing the presenter to perform various actions during the slideshow, such as starting, navigating, and ending the presentation, without the need to manually interact with the computer. This solution is particularly advantageous for scenarios where hands-free control is beneficial, such as for people with disabilities or in dynamic presentation contexts.

The primary goal of this project was to create an intuitive and user-friendly assistant that improves the presentation experience through natural gestures, minimizing disruptions and making the process more seamless and professional.

The following section documents the gestures implemented, their functionality, and the system's behavior in response to each. These gestures were designed to be intuitive and practical for real-world presentation scenarios, providing immediate feedback to the presenter through voice or visual indicators.

Implemented Intents and System Feedback

1. Start Presentation

The 'start presentation' gesture was implemented to allow presenters to initiate their slideshow seamlessly without requiring manual interaction. This feature ensures a smooth and professional start to the presentation, eliminating distractions and providing a hands-free solution.

Gesture:
Pointing up and a bit to the left, with the left arm and the hand open.

System Feedback:
The presentation and presenter view are initiated.

2. Finish Presentation

The 'end presentation' gesture was implemented to allow presenters to conclude their slideshow in a natural and elegant manner, without needing to manually interact with the computer. This gesture adds a professional touch to the presentation experience.

Gesture:
Bowing with the body.

System Feedback:
The presentation is finished and closed.

3. Next Slide

The '*next slide*' intent is one of the core functionalities of the system as it simplifies and improves the presentation flow without the need of the uses to go to the pc to change the slide. This intent was chosen because it addresses a fundamental need during presentations.

Gesture:
Raise the forearm and point your thumb to the left.

System Feedback:
The slideshow shows the next slide.

4. Previous Slide

The *'previous slide'* intent can be crucial for addressing and revisiting key points. This feature, just like the previous one, eliminates the need for physical interaction with the computer, allowing the presenter to maintain eye contact with the audience and focus on delivering their message. It ensures a seamless flow during the presentation, especially when transitioning between slides.

Example Commands:
Raise the forearm and point your thumb to the right.

System Feedback:
The slideshow shows the previous slide.

5. Advance Two Slides

The 'advance two slides' gesture was chosen to enable quick navigation during presentations, offering an efficient way to skip over less relevant slides without manual interaction, in case the presenter is running out of time, for example. This feature enhances the system's usability, catering to scenarios where the presenter needs to adjust the flow dynamically and maintain audience engagement.

Gesture:
Gesture of "two fingers" raised at the belly level, with the left hand.

System Feedback:
Success: Advances 2 slides.
Error (e.g., insufficient slides remaining): "Não é possível saltar 2 slides. Já está no final da apresentação." -> *Voice Feedback*

6. Show Elapsed Time

The 'show elapsed time' gesture provides the presenter with the time elapsed since the start of the presentation, helping them manage their time effectively and stay within the allocated schedule. This feature ensures presenters maintain control over their timing without needing to rely on external devices or tools.

Gesture:
Raising the left arm as if to check the hours and pointing to the wrist with the right hand. (as if checking a watch)

System Feedback:
“ Tempo decorrido: {elapsedTime.Hours} horas, {elapsedTime.Minutes} minutos e {elapsedTime.Seconds} segundos.” -> *Voice Feedback*

7. Request Silence

The ‘request silence’ gesture allows the presenter to politely ask the audience for silence in a non-disruptive and natural way. This feature is particularly useful in situations where the presentation includes more than one presenter and we don’t want to interrupt our colleague talking and so this is crucial to maintain focus and reducing distractions.

Gesture:
"Shhh" gesture (hand on mouth with the right hand index finger raised).

System Feedback:
“Pedimos silêncio à audiência, por favor!” -> *Voice Feedback*

8. Does anyone have questions

The ‘does anyone have questions’ gesture enables the presenter to smoothly transition into a Q&A session, engaging the audience and encouraging interaction. This gesture provides a natural and intuitive way to open the floor for questions without needing manual intervention.

Gesture:
Open your arms outstretched forward.

System Feedback:
“Alguém tem alguma dúvida?” -> *Voice Feedback*

Conclusion

The implemented gestures demonstrate the versatility and user-centric design of the gesture-controlled PowerPoint system. From basic navigation gestures to dynamic interactions like time tracking and audience engagement, the system collectively provides an intuitive and practical tool for presenters. Each gesture has been tested to ensure reliability and responsiveness, making this solution a valuable enhancement to traditional presentation tools. This project not only showcases the potential of gesture control in professional settings but also highlights its inclusive nature, serving as a foundation for further development and refinement. We learned a lot with this project and hope to improve our system in the next one with Fusion.